Worth the Tokens? A Pigeonhole Bound on Reasoning, and Why Length-Biased Judges Ignore It

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Abstract

When text is cheap, length no longer signals effort, yet reasoning is still often rewarded by how much of it there is. We prove two elementary facts. First, accuracy is bounded by one, so at most $1/c$ reasoning blocks can each be worth a gain c, for any model. Second, any judge with a positive per-token bias prefers, for a large enough budget cap, a policy that is worse than not reasoning at all. We state what neither fact implies and propose a falsifiable test.

Results. Fix a model, a task distribution, and a procedure that truncates the reasoning trace at n tokens and then forces an answer. Let $a(n) \in [0,1]$ be the expected accuracy and $\bar{a}(n) = \max_{m \le n} a(m)$ the best accuracy attainable with budget at most n ($\bar{a}$ is nondecreasing even if a is not). For block size $k \ge 1$, block j has gain $g_j = \bar{a}((j+1)k) - \bar{a}(jk) \ge 0$.

Proposition 1 (Pigeonhole)**.** *For all $c > 0$ and $k \ge 1$: $|\{j : g_j \ge c\}| \le (1 - a(0))/c \le 1/c$, and the bound is attained.*

Proof. The gains are nonnegative and telescope, $\sum_{j<J} g_j = \bar{a}(Jk) - \bar{a}(0) \le 1 - a(0)$. If M blocks had $g_j \ge c$ then $Mc \le 1 - a(0)$. Equality holds when a rises by exactly c at each of $1/c$ blocks and is then constant. □

Remark. The same counting holds in information units: since $I(Y; Z_{1:T} \mid X) = \sum_t I(Y; z_t \mid X, Z_{<t}) \le H(Y \mid X)$ for discrete Y, at most $H(Y \mid X)/\tau$ tokens carry $\ge \tau$ bits *in expectation*.

Corollary 1. *If a block of k tokens costs λ in accuracy-equivalent units, at most k/λ tokens lie in blocks worth their price.*

Proposition 2 (Goodhart)**.** *Let true utility be $u(n) = a(n) - \lambda n$ and a judge's reward $r(n) = a(n) + \beta n$, with $\lambda, \beta > 0$, over budgets $n \in \{0, \dots, N\}$. Every maximiser n^* of r satisfies $n^* > N - 1/\beta$. Hence if $N > 1/\lambda + 1/\beta$, then $u(n^*) < 0 \le u(0)$.*

Proof. $r(N) \ge \beta N$, whereas $r(n) \le 1 + \beta n < \beta N$ whenever $n < N - 1/\beta$; so no such n maximises r. Then $n^* > 1/\lambda$ gives $u(n^*) \le 1 - \lambda n^* < 0 \le a(0) = u(0)$. □

Limits. Proposition 1 bounds the *amount* of profitable reasoning, not its *location*: gains may follow a plateau with a jump, so stopping at the first block with $g_j < \lambda$ can be suboptimal. Proposition 2 is a worst case; it assumes a bias that is linear in length and does not show that real judges are so biased, which prior work documents only partly [4, 5].

Hypothesis 1. For a given model and benchmark: (a) profitable blocks form a prefix (no block with $g_j < \lambda$ precedes one with $g_{j'} \ge \lambda$); (b) an uncalibrated LLM judge's win-rate keeps rising with n after $\bar{a}$ has plateaued.

Protocol (proposed, not executed). On a verifiable benchmark, evaluate $a(n)$ by truncate-and-answer on a grid of budgets with several seeds; estimate each g_j with bootstrap intervals. (a) is refuted by one interval below λ preceding one above it; (b) by a judge win-rate that is flat or falling over the plateau. Per-instance stopping, which needs g_j predicted from the trace, is open.

Related work. Test-time scaling [1, 3]; overthinking [2]; length bias [4, 5]; length as a broken effort heuristic [6]. No empirical finding is claimed.

Conclusion. A bounded answer affords only a bounded number of valuable thoughts, and a judge that pays by the token will buy the rest. By Proposition 1, the marginal sentence is worth at most $1/c$; we stop here.

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References

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